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A finite difference method on quasi-uniform grids for the fractional boundary-layer Blasius flow

Alessandra Jannelli

Department of Mathematical and Computer Sciences, Physical Sciences and Earth Sciences, University of Messina, Viale F. Stagno D'Alcontres, 31 98166 Messina, Italy

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Abstract

In this paper, we propose a fractional formulation, in terms of the Caputo derivative, of the Blasius flow described by a non-linear two-point fractional boundary value problem on a semi-infinite interval. We develop a finite difference method on quasi-uniform grids, based on a suitable modification of the classical L1 approximation formula and show the consistency, the stability and the convergence. The numerical results confirm the theoretical ones. Comparisons with some recently proposed results are carried out to validate the accuracy of the obtained numerical results, and to show the efficiency and the reliability of the proposed numerical method.
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Keywords: Fractional Blasius flow; Nonlinear boundary value problems on semi-infinite domain; Caputo derivative; Quasi-uniform grid; Finite difference schemes

1. Introduction

In the last decades, fractional differential equations have become a useful tool to describe a lot of physical problems in many fields of the applied sciences. For this reason, they represent a relevant research area due to their wide application to a wide class of problems arising in many fields of the applied sciences: e.g. thermodynamics, astrophysics, chemical kinetics, mathematical physics, population models, thermal behaviour, fluid mechanics and many other topics. Many of these problems can be modeled by linear and nonlinear boundary value problems of fractional order (FBVPs) and, among these, many models arise on unbounded domains.

In this paper, we derive the fractional order Blasius model, governed by a FBVP on a semi-infinite domain with fractional derivative expressed in terms of the Caputo operator, and, then, an efficient numerical method for integrating it is proposed. As recently shown, the boundary-layer flow of viscous fluid could also depend on the concept of fractional derivatives. Pan et al. [28] studied the velocity distribution to the nonlinear space-fractional boundary layer equation by symmetry analysis. Mohamed et al. [13] solved fractional boundary-layer flow problems using the power-series technique. Chen et al. [9] treated time-space-dependent fractional boundary-layer flow of Maxwell fluid over an unsteady stretching surface. Mohammadein et al. [25] presented an approximate similarity solution of a fractional laminar boundary layer of viscous fluid flow over an infinite vertical plate computed by

E-mail address: ajannelli@unime.it.

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fractional power-series technique. Recently, [24] investigated the artificial boundary conditions for fractional partial differential equations on an unbounded domain.

The main aim of this paper is to derive the fractional order Blasius model and to propose a finite difference method implemented on quasi-uniform meshes for solving it. The physical problem is modeled by a third order nonlinear fractional ordinary differential equation, defined on a semi-infinite domain, that represents a guideline for several fractional problems arising in the gas flow theory. Recently, the fractional Blaisus flow model in terms of the Riemann–Liouville derivative has been proposed by Pan et al. [28] and numerically investigated by the generalized shooting method, employing the shifted Grunwald formula and classical fourth order Runge Kutta method as the iterative scheme implemented on a truncated domain. The truncated domain approach is the classical one generally used for solving BVPs and, then FBVPs on infinite and semi-infinite domains. It deals with replacing the infinite domain by a truncated finite interval considering a sufficiently large finite interval, the so-called truncated boundary, and then, on this new "fictitious point" to impose a suitable boundaries condition. This raises the question of what is a sufficiently large "finite interval" and, obviously, what consequences, in terms of accuracy, could arise if the fictitious point is not posed far enough and, moreover, many calculations should be done to accurately predict the location of the truncated boundary. To avoid the drawbacks of the truncated domain, to accurately predict the location of the truncated boundary and to reduce the computational cost, it is possible to construct quasi-uniform meshes [2,15–18,21] by using smooth, strictly monotonic functions. In this way, the whole infinite domain is taken into account in the mapping where the grid-points are located at the mid-point of each sub-interval, then the difficulty caused by numerical treatment of the last infinite sub-interval is avoided.

In this paper, we propose a smooth, strictly monotonic function to construct a quasi-uniform map and then we develop on the original semi-infinite domain an implicit finite difference method defined on a quasi-uniform grid that allows imposing the given boundary conditions exactly. The solution of the model under study, for different values of fractional order, is obtained numerically by the classical shooting approach employing the proposed numerical method on the logarithmic quasi-uniform grid. The reported numerical results point out the key advantage of the proposed numerical approach, that is: the given FBVP is solved on a semi-infinite interval by introducing a stencil that is constructed in such a way that the boundary conditions at infinity are exactly assigned. Finally, a comparison with the results obtained by the classical trapezoidal and L1 methods with the truncated boundary approach and the uniform grid is performed. The accuracy, efficiency and reliability of the proposed numerical approach are pointed out by the numerical test problem reported.

The structure of the paper is as follows: In Sec. 2, the mathematical model is derived from the laminar boundary layer flow over a semi-infinite flat plate governed by the two dimensional partial differential equations. The original problem is transformed into a fractional boundary value problem (FBVP) on an unbounded domain in terms of the Caputo derivative. In Sec. 3, the finite difference scheme is derived for the reduced FBVP. The consistency, stability and convergence of the present scheme are investigated. In Sec. 4, numerical examples and comparisons with some results found in the specialized literature are provided to verify our theoretical results and the reliability and effectiveness of the present scheme. Finally, we conclude the paper with Sec. 5.

2. Derivation of the mathematical model

The classical Blasius equation models the behaviour of a two-dimensional steady state laminar viscous flow of an incompressible fluid over a semi-infinite flat plate. It is obtained starting from the Prandtl's boundary layer equations, object of considerable interest because they represent an important simplification of the original Navier–Stokes equations and arise in various physical models of fluid mechanics. The Blasius boundary layer equation may be viewed as a special case of the Falkner-Skan equation and both belong to a class of mathematical models known as magneto-hydrodynamics (MHD) ones. These models are of strong interest in the design and analysis of power generators, pumps, accelerators, electrostatic filters, droplet, heat exchangers, reactors and the like. MHD boundary layer flows have been studied by several researchers.

In this Section, we show how we derive the fractional Blaisus model, object of our numerical investigation, starting from the Prandtl's equations. We consider the boundary layer equations for a steady incompressible flow with constant viscosity and density. In particular, in the equation of the moment, we substitute the derivative of integer order of the shear stress by a fractional order derivative in terms of the Caputo operator in order to describe non-local stress by taking into account the long-range interactions and, then, we proceed using the similarity transform technique as in [4]. We choose to use the Caputo derivative that has the advantage of properly handling

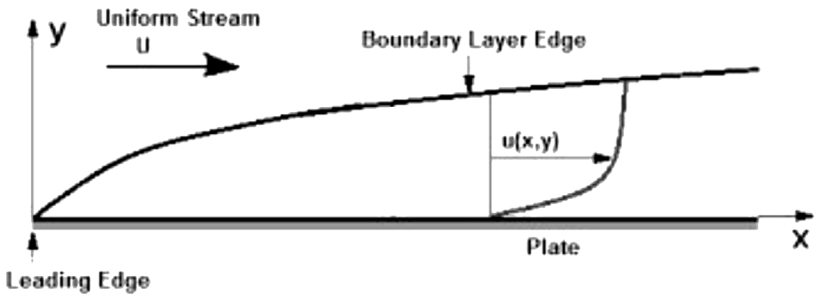

Fig. 1. Boundary layer profile.

initial value problems in which the initial conditions are given in terms of the field variables and their integer order derivatives in agreement with most of the physical processes and have a clear physical meaning.

In the framework of a Cartesian coordinate system with origin fixed at the leading edge of the vertical plate, the x-axis pointing parallel to the plate in the direction of the flow and the y-axis pointing normal to the plate, see Fig. 1, we consider the mathematical model of laminar boundary layer flow over a semi-infinite flat plate governed by the two-dimensional partial differential equations

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$
$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\frac{\partial^2 u}{\partial y^2}$$
$$\frac{\partial p}{\partial y} = 0,$$

subject to the boundary conditions

$$u(x,0) = 0 \qquad u(x,\infty) = U \qquad x > 0$$
$$u(0,y) = U \qquad y > 0 \tag{1}$$
$$v(x,0) = 0 \qquad x > 0, \tag{2}$$

with U assigned mean-stream velocity. (x, y) denote the Cartesian coordinates parallel and perpendicular to the boundary $y = 0$. $u(x, y)$ and $v(x, y)$ are the components of the velocity, the streamwise and transverse (wall normal) velocities inside the boundary layer, respectively. p is the pressure, ρ is the density and ν is the kinematic viscosity of the fluid. The boundary conditions at $y = 0$ are based on the assumption that neither slip nor mass transfer are permitted at the plate whereas the remaining boundary condition means that the velocity u tends to the main-stream velocity U asymptotically. Such flow scene is fundamental to the boundary layer theory, in which the viscous effects assume a significant role in the thin layer near the object surface. In [27,31], the main properties of the solution are discussed.

Now, we assume that the external stream is spatially uniform with speed U horizontal, that is taken parallel to the plate, such that the pressure gradient is sufficiently small as to be zero. Blasius showed that for the case where $\partial p/\partial x = 0$, the Prandtl x-momentum equation has a self-similar solution. The self-similar solution exists because the equations and the boundary conditions are invariant under a suitable transformation, see [4] for more details. In this context, the governing equations read

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$
$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = \nu\frac{\partial^2 u}{\partial y^2}. \tag{3}$$

It is well known that, in the second equation of (3), the term at the right hand side comes from the relationship between the strain rate and the shear stress for the Newtonian fluids expressed by

$$\tau_{yx} = \mu\frac{\partial u}{\partial y}$$

where μ is the dynamic viscosity. Then, deriving respect to y and dividing by ρ, we obtain the term

$$\frac{1}{\rho}\frac{\partial \tau_{yx}}{\partial y} = \frac{\mu}{\rho}\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right) = \nu\frac{\partial^2 u}{\partial y^2}, \tag{4}$$

where we set $\nu = \frac{\mu}{\rho}$.

In this paper, we follow the innovative ideas initially proposed for solid mechanics [7] and apply them to fluid mechanics: the stress of a material is defined as the fractional integral of the strain field. In classical elastic models of solid mechanics, the stress state at a given point is regarded as being determined uniquely by the strain state at that same point [29]. In [6,10,22], the spatial fractional order calculus formalism was introduced to describe non local stress by taking into account the long range interactions of the materials. For viscoelastic fluid in the boundary layer, the corresponding assumption is that fluids in a given layer are influenced by every layer from the plate to the particular layer. Then, the above relationship (4) between shear stress and strain rate is replaced with the following

$$\nu\frac{\partial^\alpha}{\partial y^\alpha}\left(\frac{\partial u}{\partial y}\right) \tag{5}$$

where $\frac{\partial^\alpha}{\partial y^\alpha}$ denotes the α-order fractional derivative of Caputo type. For a function $z(y)$ of C^1 the Caputo derivative of α order, with $0 < \alpha < 1$, is defined as follows

$$\frac{\partial^\alpha z}{\partial y^\alpha} = \frac{1}{\Gamma(1-\alpha)}\int_0^y \frac{dz}{dy}(s)(y-s)^{-\alpha}ds \tag{6}$$

with $\Gamma(\cdot)$ Euler's gamma function. In this formulation, we use the Caputo definition of the fractional derivative because, in general, it allows defining a model whose initial and boundary conditions are given in terms of the field variable and of its integer order derivatives, in agreement with the clear physical meaning of most of the processes arising in the real world.

Substituting (5) into (3) leads to the following system where the momentum equation involves the spatial fractional Caputo derivative

$$\begin{aligned} &\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \\ &u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = \nu\frac{\partial^\alpha}{\partial y^\alpha}\left(\frac{\partial u}{\partial y}\right). \end{aligned} \tag{7}$$

Note that, for $\alpha \to 1$, the classical physical model (3) is recovered. For different choices of α, the classical physical model is not recovered.

Now, we introduce a similarity variable η

$$\eta = \eta(x, y) = \frac{y}{g(x)}, \tag{8}$$

where $g(x)$ is a function that has to be found, and set u as follows

$$u(x, y) = U\ h(\eta(x, y)) = U\ h\left(\frac{y}{g(x)}\right) \tag{9}$$

with $h(\eta)$ new function of the similarity variable η. By the first of (7), we can write

$$\frac{\partial v}{\partial y} = -\frac{\partial u}{\partial x} = \frac{U\ g'(x)}{g(x)}h'(\eta)\eta.$$

Here and in the following, the $'$ denotes the derivative of the functions of one independent variable. In order to compute the explicit form of the function $v(x, y)$, we integrate the above equation with respect to y

$$\begin{aligned} v(x, y) &= \frac{U\ g'(x)}{g(x)}\int_0^y h'(\eta(x, y))\ \eta(x, y)\ dy \\ &= U\ g'(x)\int_0^\eta h'(\eta)\ \eta d\eta \end{aligned}$$

where we use the condition $v(x, 0) = 0$ for $x > 0$ and $dy = g(x)d\eta$. Then, we have

$$v(x, y) = U\ g'(x)\left(\eta\ h(\eta) - \int_0^\eta h(\eta)\ d\eta\right).$$

Now, by assuming

$$f(\eta) = \int h(\eta)\, d\eta \tag{10}$$

with $f(\eta)$ primitive function of $h(\eta)$, we obtain $f'(\eta) = h(\eta)$ so that

$$v(x, y) = U\ g'(x)\left(\eta\ f'(\eta) - f(\eta)\right)$$
$$u(x, y) = U\ h(\eta) = Uf'(\eta).$$

Finally, we have to compute

$$\begin{aligned}\frac{\partial^\alpha}{\partial y^\alpha}\left(\frac{\partial u}{\partial y}\right) &= \frac{\partial^\alpha}{\partial y^\alpha}\left(\frac{\partial}{\partial y}\left(Uf'(\eta)\right)\right)\\ &= \frac{\partial^\alpha}{\partial y^\alpha}\left(\frac{U}{g(x)}f''(\eta)\right)\\ &= \frac{U}{g(x)}\frac{\partial^\alpha}{\partial y^\alpha}\left(f''(\eta)\right)\end{aligned}$$

In order to compute

$$\frac{\partial^\alpha}{\partial y^\alpha}\left(f''(\eta)\right),$$

we recall the definition of Caputo fractional derivative of the function f'' with respect to another function $y = y(\eta) = \eta g(x)$ (see definition (1) in [1] for $\alpha \in (0, 1)$)

$$\begin{aligned}\frac{\partial^\alpha}{\partial y^\alpha}\left(f''(\eta)\right) &= \frac{1}{\Gamma(1-\alpha)}\int_0^\eta y'(\tau)\ (y(\eta) - y(\tau))^{-\alpha} f''_y(\tau)d\tau\\ &= \frac{1}{\Gamma(1-\alpha)}\int_0^\eta y'(\tau)\ (g(x)\eta - g(x)\tau)^{-\alpha} f''_y(\tau)d\tau\\ &= g(x)^{-\alpha}\frac{1}{\Gamma(1-\alpha)}\int_0^\eta (\eta-\tau)^{-\alpha} f'''(\tau)d\tau\\ &= g(x)^{-\alpha} D_0^\alpha f''(\eta)\end{aligned}$$

being

$$D_0^\alpha f''(\eta) = \frac{1}{\Gamma(1-\alpha)}\int_0^\eta (\eta-\tau)^{-\alpha} f'''(\tau)d\tau \tag{11}$$

the classical fractional order Caputo derivative for a function of a single variable.

By suitable substitutions, we obtain

$$D_0^\alpha f''(\eta) + \frac{U}{\nu}g'(x)g^\alpha(x)\ f(\eta)f''(\eta) = 0.$$

Now, we want that the above equation becomes the classical Blasius equation for $\alpha = 1$, that is, becomes an ordinary differential equation in the only unknown $f(\eta)$. This implies that the coefficient containing the function $g(x)$ must be constant. In this context, the function $g(x)$, which allows to determine a similar solution of the problem, has to be determined as the solution of the equation

$$\frac{U}{\nu}g'(x)g^\alpha(x) = \frac{1}{1+\alpha}.$$

We get

$$g(x) = \left(\frac{\nu x}{U}\right)^{\frac{1}{1+\alpha}}.$$

Then, the similarity variable is defined as follows

$$\eta = \frac{y}{g(x)} = y\left(\frac{U}{\nu x}\right)^{\frac{1}{1+\alpha}} = \frac{y}{x}\left(\frac{Ux^\alpha}{\nu}\right)^{\frac{1}{1+\alpha}}$$

and the fractional Blasius model reads

$$D_0^\alpha f''(\eta) + \frac{1}{1+\alpha}\ f(\eta)\ f''(\eta) = 0 \qquad \eta \in [0, \infty) \tag{12}$$

whose the boundary conditions are obtained by (1)

$$f(0) = f'(0) = 0 \qquad \lim_{\eta\to\infty} f'(\eta) = 1\ . \tag{13}$$

When $\alpha = 1$, Eq. (12) collapses into the following classical well known Blasius equation

$$f'''(\eta) + \frac{1}{2} f(\eta) f''(\eta) = 0, \qquad \eta \in [0, \infty) \tag{14}$$

derived by Blasius using a similarity transform technique [4]. It represents the simplest nonlinear boundary-layer problem arising when studying a laminar boundary-layer problem for Newtonian fluids. The flow is usually called the boundary layer flow since the viscous effects are limited to a thin layer near the flat plate surface. The investigation of Blasius problem has developed a vast bibliography: the model (14) has been analytically and numerically investigated by several authors and different techniques have been developed to obtain analytical and numerical solutions. Note that, the solution $f(\eta)$ of the classical Blasius model is defined in terms of the solution $u(x, y)$ of the Prandatl system, Then, the regularity property of the Blasius model results by the regularity of the solution of the Prandatl model. The integer order model is studied intensively in the literature. To the author's knowledge, not much research has been done on fractional order boundary layer models.

Finally, we define the model (12) with boundary conditions (13) as the fractional boundary value problem (FBVP), object of our numerical study reported in the next Section.

3. Quasi-uniform grid and numerical method

In this Section, we present the finite difference method developed in order to approximate the solution of the given FBVP. As usual, in order to solve a BVP, it is recovered as an initial value problem and, then, integrated by a IVP solvers. In this context, we proceed following the same way and proposing as IVP solver a new finite difference method. Of the proposed numerical method, we discuss its properties following the classical analysis for IVP solvers.

The numerical method, proposed for the problem under study, takes into account not only the asymptotic boundary conditions but also the behaviour of the solution at the initial point. We consider the discretization of the (12)–(13) on a quasi-uniform mesh, where the Caputo derivative is approximated by a suitable modification of the popular L1 formula and, then, a finite difference operator is obtained. Our interest here is to show how the layer in the solution at the initial point is treated and how, by the proposed numerical method, we overcome the drawback of the boundary conditions imposed at infinity.

The model (12) is an example of multi-order fractional equation, then, at the first step, we set

$$u_1 = f(\eta) \qquad u_2 = f'(\eta) \qquad u_3 = f''(\eta)$$

and rewrite the FBVP (12)–(13) as a system of fractional nonlinear ordinary differential equations [11]

$$\begin{aligned} u_1' &= u_2 \\ u_2' &= u_3 \\ D_0^\alpha u_3 &= -\frac{1}{1+\alpha} u_1 u_3 \\ u_1(0) &= u_2(0) = 0\ , \qquad \lim_{\eta\to\infty} u_2(\eta) = 1, \end{aligned} \tag{15}$$

where $D_0^\alpha u_3$ is the α order Caputo derivative of $u_3 = f''(\eta)$ defined by (11). To solve the FBVP (15) on the semi-infinite domain, first, we construct a quasi-uniform map from a referenced finite domain and then, we develop on the original domain a finite difference scheme that allows imposing the given boundary conditions exactly.

Several kinds of maps can be used based on the behaviour of the solution. Because it is well known that the solution of the Blasius flow is shown to have an initial layer at the beginning of the process, for small values of η, it is expected that better results are obtained by using of the following quasi-uniform map $\eta = \eta(\xi)$,

$$\eta = -c \cdot \ln(1 - \xi)\ , \tag{16}$$

Fig. 2. Quasi-uniform mesh (16). We notice that the last mesh-point is $\eta_N = \infty$.

where $\xi \in [0, 1]$, $\eta \in [0, \infty)$, and $c > 0$ is a control parameter that defines the grid points distribution in the original physical infinite domain: a denser grid points distribution near the initial point and, then, a coarse grid point one with an increasing spatial resolution going towards infinity.

By (16), a family of uniform grids $\xi_n = n/N$ defined on finite interval $[0, 1]$ generates a one-parameter family of quasi-uniform grids $\eta_n = \eta(\xi_n)$ on the semi-infinite interval $[0, \infty)$. The FBVP (15) is discretized by introducing a uniform grid ξ_n of $N + 1$ nodes in $[0, 1]$ such that $\xi_0 = 0$ and $\xi_{n+1} = \xi_n + h$ with $h = 1/N$. In this way, η_n is a quasi-uniform grid in $[0, \infty)$. The last interval in (16), namely $[\eta_{N-1}, \eta_N]$, is infinite but the mid-point $\eta_{N-1/2}$ is finite, because the non integer nodes of the map are defined by $\eta_{n+j} = \eta\left(\xi = \frac{n+j}{N}\right)$, with $n \in \{0, 1, \ldots, N-1\}$ and $0 \le j \le 1$. This map allows describing the infinite domain by a finite number of intervals, with the last infinite interval being the last node placed at infinity, $\eta_N = \eta(\xi_N) = \infty$. In this way, the boundary conditions are assigned exactly.

Fig. 2 shows an example of quasi-uniform mesh $\eta = \eta_n$, $n = 0, 1, \ldots, N$ defined by (16) with $c = 5$ and $N = 20$. Note that the map (16) generates a monotonically increasing sequence of time steps, $\Delta\eta_n < \Delta\eta_{n+1}$, with $\Delta\eta_n = \eta_n - \eta_{n-1}$ and $\Delta\eta_{n+1} = \eta_{n+1} - \eta_n$ $\forall n = 1, \ldots, N-1$. In this context, the logarithmic map (16) reveals an efficient grid because it gives a better resolution near to $\eta = 0$, point where the solution of the problem under study reveals a layer.

For sake of simplicity of exposition, in the following, we consider a scalar function $u(\eta)$. In order to construct a numerical method that uses the value of the solution at infinity, that is $u(\eta_N) = u(\infty)$, and not the point mesh $\eta_N = \infty$, we approximate the values of the scalar variable $u(\eta_{n+1/2})$ and its first derivative at mid-points of the grid, $\eta_{n+1/2}$ for $n = 0, \ldots, N-1$, using the following difference discretization formulae

$$u(\eta_{n+\frac{1}{2}}) \approx \frac{\eta_{n+\frac{3}{4}} - \eta_{n+\frac{1}{2}}}{\eta_{n+\frac{3}{4}} - \eta_{n+\frac{1}{4}}}\, u(\eta_n) \;+\; \frac{\eta_{n+\frac{1}{2}} - \eta_{n+\frac{1}{4}}}{\eta_{n+\frac{3}{4}} - \eta_{n+\frac{1}{4}}}\, u(\eta_{n+1}) \tag{17}$$

$$u'(\eta_{n+\frac{1}{2}}) \approx \frac{u(\eta_{n+1}) - u(\eta_n)}{2(\eta_{n+\frac{3}{4}} - \eta_{n+\frac{1}{4}})}. \tag{18}$$

To get (17) we can apply the Taylor formula, at $\eta_{n+\frac{1}{2}}$, to both $u(\eta_{n+1})$ and $u(\eta_n)$. A simpler way to obtain (17) is to consider the method of the undefined coefficients, for $u(\eta_{n+\frac{1}{2}})$ as a linear combination of $u(\eta_n)$ and $u(\eta_{n+1})$, and to require that the formula is exact for constant and linear functions. In this way, we end up with a linear system of two equations in two unknowns where the coefficient matrix is a Vandermonde matrix. Furthermore, in order to justify non-standard finite difference formula (18), we note that, by considering $u = u(\xi(\eta))$, we can write

$$\left.\frac{du}{d\eta}\right|_{n+\frac{1}{2}} = \left.\frac{du}{d\xi}\right|_{n+1/2} \left.\frac{d\xi}{d\eta}\right|_{n+\frac{1}{2}} \approx \frac{u_{n+1} - u_n}{\xi_{n+1} - \xi_n} \frac{2\left(\xi_{n+\frac{3}{4}} - \xi_{n+\frac{1}{4}}\right)}{2\left(\eta_{n+\frac{3}{4}} - \eta_{n+\frac{1}{4}}\right)}\,. \tag{19}$$

The right hand side of Eq. (19) reduces to the right hand side of Eq. (18) because we are using a uniform grid for ξ and therefore $2\left(\xi_{n+3/4} - \xi_{n+1/4}\right) = \xi_{n+1} - \xi_n$. The two finite difference approximations (17) and (18) have order of accuracy $O(N^{-2})$. For a system of differential equations, they can be applied component-wise. The approximation formula (17) and (18) were proposed in [16], and successively adopted in [17,18], for solving the class of integer order BVPs defined on semi-infinite domain. A first version of the discretization formula for solving BVPs was proposed in [15].

In Fig. 3, we report a sample of the generic interval $[\eta_n, \eta_{n+1}]$, for $n = 0, 1, \ldots, N-1$, of the proposed quasi-uniform grid, where for the subintervals the following relation is verified

$$(\eta_{n+\frac{1}{4}} - \eta_n) < (\eta_{n+\frac{1}{2}} - \eta_{n+\frac{1}{4}}) < (\eta_{n+\frac{3}{4}} - \eta_{n+\frac{1}{2}}) < (\eta_{n+1} - \eta_{n+\frac{3}{4}}).$$

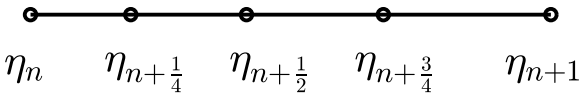

Fig. 3. Quasi-uniform mesh (16): An example of interval $[\eta_n, \eta_{n+1}]$.

Now, we propose a suitable approximation formula for the Caputo derivative of $u(\eta)$ at mid-points of the grid $\eta_{n+1/2}$. We recall the following classical formula, the so-called L1 formula [26], to approximate the Caputo derivative of the scalar variable $u(\eta)$ evaluated at mesh point η_{n+1}

$$\begin{aligned} D_0^\alpha u(\eta_{n+1}) &= \frac{1}{\Gamma(1-\alpha)} \int_0^{\eta_{n+1}} u'(s)(\eta_{n+1}-s)^{-\alpha} ds \\ &= \frac{1}{\Gamma(1-\alpha)} \int_0^{\eta_{n+1}} u'(\eta_{n+1})(\eta_{n+1}-s)^{-\alpha} ds + R_{n+1} \\ &= \frac{1}{\Gamma(1-\alpha)} \sum_{k=0}^{n} \frac{u_{k+1}-u_k}{\eta_{k+1}-\eta_k} \int_{\eta_k}^{\eta_{k+1}} (\eta_{n+1}-s)^{-\alpha} ds + R_{n+1}, \end{aligned} \tag{20}$$

where R_{n+1} is the local error. For $0 < \alpha < 1$ and for a function $f(\eta) \in C^2([a,b])$ it holds that $R_{n+1} = O((\Delta\eta)^{2-\alpha})$, as shown by Oldham in [26] on uniform grids with constant step size $\Delta\eta$. We obtain the following approximation

$$D_\eta^\alpha u(\eta_{n+1}) \approx \frac{1}{\Gamma(2-\alpha)} \sum_{k=0}^{n} T_{n+1,k}\ (u_{k+1}-u_k) \tag{21}$$

where

$$T_{n+1,k} = \frac{(\eta_{n+1}-\eta_k)^{1-\alpha} - (\eta_{n+1}-\eta_{k+1})^{1-\alpha}}{\eta_{k+1}-\eta_k}$$

being

$$\int_{\eta_k}^{\eta_{k+1}} (\eta_{n+1}-s)^{-\alpha} ds = \frac{1}{1-\alpha}\left((\eta_{n+1}-\eta_k)^{1-\alpha} - (\eta_{n+1}-\eta_{k+1})^{1-\alpha}\right).$$

In order to construct the proposed method, now, we evaluate the formula (20) not at the mesh point η_{n+1}, but at the mesh mid-point $\eta_{n+\frac{1}{2}}$

$$\begin{aligned} D_\eta^\alpha u(\eta_{n+\frac{1}{2}}) &= \frac{1}{\Gamma(1-\alpha)} \int_0^{\eta_{n+\frac{1}{2}}} u'(s)(\eta_{n+\frac{1}{2}}-s)^{-\alpha} ds \\ &\approx \frac{1}{\Gamma(1-\alpha)} \int_0^{\eta_{n+\frac{1}{2}}} u'(\eta_{n+\frac{1}{2}})\ (\eta_{n+\frac{1}{2}}-s)^{-\alpha} ds \\ &\approx \frac{1}{\Gamma(1-\alpha)} \sum_{k=0}^{n} \frac{u_{k+1}-u_k}{2(\eta_{k+\frac{3}{4}}-\eta_{k+\frac{1}{4}})} \int_{\eta_{k-\frac{1}{2}}}^{\eta_{k+\frac{1}{2}}} (\eta_{n+\frac{1}{2}}-s)^{-\alpha} ds \end{aligned}$$

where the first derivative $u'(\eta_{n+\frac{1}{2}})$ is approximated by (17). Note that, we introduce a fictitious point for $n = 0$, namely $\eta_{-\frac{1}{2}}$. Here, we use u_n and u_{n+j}, with $0 \le j \le 1$, as numerical approximations of the function $u(\eta_n)$ and $u(\eta_{n+j})$ at the mesh points η_n and η_{n+j}, respectively. Finally, we obtain the following formula

$$D_\eta^\alpha u(\eta_{n+\frac{1}{2}}) \approx \frac{1}{\Gamma(2-\alpha)} \sum_{k=0}^{n} T_{n+\frac{1}{2},k}\ (u_{k+1}-u_k) \tag{22}$$

where

$$T_{n+\frac{1}{2},k} = \frac{(\eta_{n+\frac{1}{2}}-\eta_{k-\frac{1}{2}})^{1-\alpha} - (\eta_{n+\frac{1}{2}}-\eta_{k+\frac{1}{2}})^{1-\alpha}}{2(\eta_{k+\frac{3}{4}}-\eta_{k+\frac{1}{4}})}$$

being

$$\int_{\eta_{k-\frac{1}{2}}}^{\eta_{k+\frac{1}{2}}} (\eta_{n+\frac{1}{2}} - s)^{-\alpha} ds = \frac{1}{1-\alpha} \left((\eta_{n+\frac{1}{2}} - \eta_{k-\frac{1}{2}})^{1-\alpha} - (\eta_{n+\frac{1}{2}} - \eta_{k+\frac{1}{2}})^{1-\alpha} \right).$$

By using the formulae (17),(18) and (22) for the three components of the solution ${}^{\ell}u$, for $\ell = 1, 2, 3$, we obtain the following implicit finite difference scheme for the model under study

$$\begin{aligned} {}^{1}u_{n+1} &= {}^{1}u_n + h_0\, {}^{2}u_{n+\frac{1}{2}} \\ {}^{2}u_{n+1} &= {}^{2}u_n + h_0\, {}^{3}u_{n+\frac{1}{2}} \\ {}^{3}u_{n+1} &= {}^{3}u_n - \frac{h_0}{h_1} \left(\sum_{k=0}^{n-1} T_{n+\frac{1}{2},k} \left({}^{3}u_{k+1} - {}^{3}u_k \right) + \frac{\Gamma(2-\alpha)}{1+\alpha}\, {}^{1}u_{n+\frac{1}{2}}\, {}^{3}u_{n+\frac{1}{2}} \right) \end{aligned} \tag{23}$$

for $n = 0, \ldots, N-1$, where

$${}^{l}u_{n+\frac{1}{2}} = h_n\, {}^{l}u_n + h_{n_1}\, {}^{l}u_{n+1}, \qquad l = 1, 2, 3, \tag{24}$$

with

$$h_n = \frac{\eta_{n+\frac{3}{4}} - \eta_{n+\frac{1}{2}}}{\eta_{n+\frac{3}{4}} - \eta_{n+\frac{1}{4}}} \quad h_{n_1} = \frac{\eta_{n+\frac{1}{2}} - \eta_{n+\frac{1}{4}}}{\eta_{n+\frac{3}{4}} - \eta_{n+\frac{1}{4}}}.$$

and where we take into account that

$$T_{n+\frac{1}{2},n} = \frac{(\eta_{n+\frac{1}{2}} - \eta_{n-\frac{1}{2}})^{1-\alpha}}{2(\eta_{n+\frac{3}{4}} - \eta_{n+\frac{1}{4}})}$$

with

$$h_0 = 2(\eta_{n+\frac{3}{4}} - \eta_{n+\frac{1}{4}}) \qquad h_1 = (\eta_{n+\frac{1}{2}} - \eta_{n-\frac{1}{2}})^{1-\alpha}.$$

We use the notation by Lambert in order to denote ${}^{\ell}u$ the components of the solution for $\ell = 1, 2, 3$ and, moreover, assume the convention that the summation in (23) is equal to zero if the lower bound is larger than the upper bound. In compact form, the proposed numerical method can be written as follows

$$\begin{aligned} &\mathbf{u}_{n+1} - \mathbf{u}_n - h\, \mathbf{g}\left(\eta_{n+1/2}, \mathbf{u}_n, \mathbf{u}_{n+1}\right) = \mathbf{0}\ , \\ &\mathbf{B}c\left(\mathbf{u}_0, \mathbf{u}_N\right) = \mathbf{0}\ , \end{aligned} \tag{25}$$

for $n = 0, \ldots, N-1$, where $h = \frac{h_0}{h_1}$ and with $\mathbf{B}c\left(\mathbf{u}_0, \mathbf{u}_N\right) = ({}^{1}u_0, {}^{2}u_0, {}^{2}u_N - 1)^T$. The (25) represents a FBVP governed by a system of 3 nonlinear equations in the 3 unknowns.

We remark that the key advantage of the proposed finite difference formulation is to overcome the difficulty of the numerical treatment of the boundary conditions at the infinity: the formulae (17),(18) and (22), implemented to construct the numerical method (23), use the value $u(\eta_N) = u(\infty)$ and not the grid point $\eta_N = \infty$ that, in general, cannot be treated from a numerical point of view. In this way, the boundary conditions at infinity are taken into account in a natural way. Furthermore, the problem (12)–(13) is shown to have an initial layer near the initial point $\eta = 0$: the proposed finite difference method implemented on the logarithmic quasi-uniform grid takes into account the behaviour of the solution at the initial point.

3.1. Consistency, stability, and convergence

Here, our object is to show that the developed method is a reliable and efficient method for the problem under study. For this aim, in this Section, we prove the properties of consistency, zero-stability and convergence of the proposed method for solving the nonlinear model under study.

The following analysis is performed considering the regularity of solution $u(\eta) \in C^2([a, b])$ of the fractional Blasius model.

Consistency

According to (17) and (18), for the first two equations of (23), we have

$$\begin{aligned}
{}^1R_{n+\frac{1}{2}} &= \frac{{}^1u(\eta_{n+1}) - {}^1u(\eta_n)}{h_0} - {}^2u(\eta_{n+\frac{1}{2}}) \\
&= \frac{{}^1u(\eta_{n+1}) - {}^1u(\eta_n)}{h_0} - {}^2u(\eta_{n+\frac{1}{2}}) - {}^1u'(\eta_{n+\frac{1}{2}}) + {}^2u(\eta_{n+\frac{1}{2}}) \\
&= O(N^{-2})
\end{aligned}$$

$$\begin{aligned}
{}^2R_{n+\frac{1}{2}} &= \frac{{}^2u(\eta_{n+1}) - {}^2u(\eta_n)}{h_0} - {}^3u(\eta_{n+\frac{1}{2}}) \\
&= \frac{{}^2u(\eta_{n+1}) - {}^2u(\eta_n)}{h_0} - {}^3u(\eta_{n+\frac{1}{2}}) - {}^2u'(\eta_{n+\frac{1}{2}}) + {}^3u(\eta_{n+\frac{1}{2}}) \\
&= O(N^{-2}).
\end{aligned}$$

Then, the first two equations of (23) are consistent with the model under study of order $O(N^{-2})$. According to (22), it has

$$\begin{aligned}
{}^3R_{n+\frac{1}{2}} &= \frac{1}{\Gamma(2-\alpha)} \sum_{k=0}^{n} T_{n+1,k} \left({}^3u(\eta_{k+1}) - {}^3u(\eta_k) \right) + \frac{1}{1+\alpha} \, {}^1u(\eta_{n+\frac{1}{2}}) \, {}^3u(\eta_{n+\frac{1}{2}}) \\
&= \frac{1}{\Gamma(2-\alpha)} \sum_{k=0}^{n} T_{n+1,k} \left({}^3u(\eta_{k+1}) - {}^3u(\eta_k) \right) + \frac{1}{1+\alpha} \, {}^1u(\eta_{n+\frac{1}{2}}) \, {}^3u(\eta_{n+\frac{1}{2}}) \\
&\quad - D_\eta^\alpha \, {}^3u(\eta_{n+\frac{1}{2}}) - \frac{1}{1+\alpha} \, {}^1u(\eta_{n+\frac{1}{2}}) \, {}^3u(\eta_{n+\frac{1}{2}}) \\
&= O(N^{\alpha-2}).
\end{aligned}$$

As shown in [8] [Theor. 4.2 and Theor. 4.3] for $0 < \alpha < 1$ and for a function $f(\eta) \in C^2([a,b])$, on quasi-uniform grids with non constant step size $\Delta\eta$, the local truncation error of the third equation of the finite difference scheme (23) is of the $O(N^{-(2-\alpha)})$ order. Then we can conclude that the proposed method, under suitable regularity of the solution, is consistent of order $O(N^{\alpha-2})$.

Stability

In order to show the stability property of the proposed method, we perform the same approach that, as usual, is used for the numerical methods for nonlinear systems of ODEs of integer order [3,23]. In fact, "the best way to set out our objectives for a good numerical scheme for fractional differential equations is to base them on the well-established desirable characteristics of solution methods for ordinary (integer-order) differential equations", see [12].

We consider the stability of the numerical method with respect to "small perturbations" in the starting conditions. Let ${}^l\bar{u}_n$ be another numerical solution obtained by the scheme (23) with initial conditions ${}^l\bar{u}_0$. We report the following definition, [30]:

Definition. A numerical method is said to be zero-stable if there exists a quantity C_α, namely constant of stability, independent of discretization step, such that, for any integer n and for any two sequences u_n and $\bar{u}_n$ that have been generated by the same formulae but different starting values, u_0 and $\bar{u}_0$ respectively, we have

$$|u_n - \bar{u}_n| \le C_\alpha \; |u_0 - \bar{u}_0| \, ,$$

as the time step tends to zero.

In other words, the distance between the two numerical solutions based on two distinct initial conditions remains bounded by a suitable constant. This notion of stability refers to the stability of the numerical method with respect to "small perturbations" in the starting conditions.

Let ${}^l\rho_n$ be the corresponding round-off errors defined as follows

$$ {}^l\rho_n = {}^l u_n - {}^1\bar{u}_n \ , \qquad 0 \le n \le N, \qquad l = 1, 2, 3. $$

In order to demonstrate whether the finite difference scheme is stable, we study how the size of the round-off errors ${}^l\rho_n$ evolve in time,

$$ |\,{}^l\rho_n| \le C_\alpha \ |\,{}^l\rho_0| \ , \qquad\qquad l = 1, 2, 3. $$

The round-off errors satisfy the following round-off equations

$$ \begin{aligned} {}^1\rho_{n+1} &= {}^1\rho_n \\ {}^2\rho_{n+1} &= {}^2\rho_n \\ {}^3\rho_{n+1} &= {}^3\rho_n - \frac{h_0}{h_1}\left(\sum_{k=0}^{n-1} T_{n+\frac{1}{2},k}\ ({}^3\rho_{k+1} - {}^3\rho_k) + \frac{\Gamma(2-\alpha)}{1+\alpha}\ {}^1u\ {}^3\rho_{n+\frac{1}{2}}\right). \end{aligned} \tag{26} $$

Note that, as usual in the stability analysis, the terms at the right hand sides in the (23), namely ${}^2u_{n+\frac{1}{2}}$, ${}^3u_{n+\frac{1}{2}}$ and ${}^1u_{n+\frac{1}{2}}$, are assumed to be constant. Then, in the first two equations of (26), for $l = 1, 2$, it easily verifies that $|{}^l\rho_{n+1}| = |\,{}^l\rho_0|$. A more detailed analysis has to be performed for the third equation of (26) where, we set, at each step, ${}^1u = \max_{0\le n\le N} |{}^1u_{n+\frac{1}{2}}|$. Moreover, taking into account that

$$ \begin{aligned} &\sum_{k=0}^{n-1} T_{n+\frac{1}{2},k}\ ({}^3\rho_{k+1} - {}^3\rho_k) = \\ &\sum_{k=0}^{n-1} {}^3\rho_{k+1}(T_{n+\frac{1}{2},k} - T_{n+\frac{1}{2},k+1}) - T_{n+\frac{1}{2},0}\ {}^3\rho_0 \ + \ T_{n+\frac{1}{2},n}\ {}^3\rho_n, \end{aligned} $$

we can write

$$ \begin{aligned} {}^3\rho_{n+1} = &\left(1 + \frac{h_0}{h_1}\frac{\Gamma(2-\alpha)\ {}^1u\ h_{n_1}}{1+\alpha}\right)^{-1}\left(\left(1 - \frac{h_0}{h_1}\frac{\Gamma(2-\alpha)\ {}^1u\ h_n}{1+\alpha}\right)\ {}^3\rho_n\right. \\ &\left. + \ \frac{h_0}{h_1}\left(\sum_{k=0}^{n-1} {}^3\rho_{k+1}(T_{n+\frac{1}{2},k+1} - T_{n+\frac{1}{2},k}) + T_{n+\frac{1}{2},0}\ {}^3\rho_0 - T_{n+\frac{1}{2},n}\ {}^3\rho_n\right)\right). \end{aligned} $$

By assuming

$$ |{}^3\bar{\rho}_n| = \max_{0\le k\le n} |\,{}^3\rho_k| \ , $$

and being $\sum_{k=0}^{n-1}(T_{n+\frac{1}{2},k+1} - T_{n+\frac{1}{2},k}) + T_{n+\frac{1}{2},0} - T_{n+\frac{1}{2},n} = 0$, we obtain

$$ \left|{}^3\rho_{n+1}\right| \le K\ |{}^3\rho_n|, $$

where

$$ K = \left|\frac{h_1 - c_0\ h_0 h_n}{h_1 + c_0 h_0 h_{n_1}}\right| \tag{27} $$

with $c_0 = \frac{\Gamma(2-\alpha)\ {}^1u}{1+\alpha}$. Now, we have to find an upper bound for K independent of step size, then, we write

$$ K \le \left|\frac{h_1 - c_0\ h_0 h_n}{h_1}\right| \le \left|1 - c_0\frac{h_0 h_n}{h_1}\right| \le 1 + \left|c_0\frac{h_0 h_n}{h_1}\right|. $$

It easily verifies that

$$ \left|c_0\frac{h_0 h_n}{h_1}\right| = c_0\ \frac{2(\eta_{n+\frac{3}{4}} - \eta_{n+\frac{1}{2}})}{(\eta_{n+\frac{1}{2}} - \eta_{n-\frac{1}{2}})^{1-\alpha}} \le c_0\frac{\Delta\eta_n}{\Delta\eta_{n-1}^{1-\alpha}} \le c_0\frac{\Delta\eta_{n+1}}{\eta_1^{1-\alpha}} $$

with $\eta_1 = \Delta\eta_1 < \cdots < \Delta\eta_{n-1} < \eta_{n+\frac{1}{2}} - \eta_{n-\frac{1}{2}} < \Delta\eta_n < \Delta\eta_{n+1}$. Then, observing that $1 + x < e^x$, with $x > 0$, we have

$$K \leq 1 + c_0 \frac{\Delta\eta_{n+1}}{\eta_1^{1-\alpha}} \leq e^{c_0 \frac{\Delta\eta_{n+1}}{\eta_1^{1-\alpha}}}.$$

Then, we can write

$$\left|{}^3\rho_{n+1}\right| \leq e^{c_0 \frac{\Delta\eta_{n+1}}{\eta_1^{1-\alpha}}} \, |{}^3\rho_n|,$$

and, then

$$\left|{}^3\rho_{n+1}\right| \leq e^{c_0(n+1) \frac{\Delta\eta_{n+1}}{\eta_1^{1-\alpha}}} \, |{}^3\rho_0|.$$

Now, being the grid quasi-uniform, we observe that $(n+1)\Delta\eta_{n+1} \neq (\eta_{n+1} - \eta_0)$, but $(n+1)\Delta\eta_{n+1} = \tilde{\eta}$ with $\tilde{\eta} \in [0, \eta_N)$ and $\tilde{\eta} < \eta_N$. Then, we can write

$$e^{c_0(n+1) \frac{\Delta\eta_{n+1}}{\eta_1^{1-\alpha}}} = e^{c_0 \frac{\tilde{\eta}}{\eta_1^{1-\alpha}}}.$$

The following result is obtained

$$\left|{}^3\rho_{n+1}\right| \leq C_\alpha \, |{}^3\rho_0|,$$

with $C_\alpha = e^{c_0 \frac{\tilde{\eta}}{\eta_1^{1-\alpha}}}$, constant independent of the step size. We conclude that the present method is zero-stable.

Convergence

As before, to show the convergence of the proposed method, we consider the third equation of (23) written in the following form

$$\begin{aligned} {}^3u_{n+1} = {} & \frac{\left(1 - c_0 \frac{h_0 h_n}{h_1}\right)}{\left(1 + c_0 \frac{h_0 h_{n_1}}{h_1}\right)} \, {}^3u_n - \\ & \frac{h_0}{h_1} \frac{1}{\left(1 + c_0 \frac{h_0 h_{n_1}}{h_1}\right)} \left(\sum_{k=0}^{n-1} {}^3u_{k+1} (T_{n+\frac{1}{2},k+1} - T_{n+\frac{1}{2},k}) + T_{n+\frac{1}{2},0} \, {}^3u_0 - T_{n+\frac{1}{2},n} \, {}^3u_n \right). \end{aligned}$$

Let ${}^3u(\eta_n)$ be the exact solution of the above equation at mesh point η_n for $n = 0, 1, \ldots, N-1$. We denote by ${}^3\epsilon_n = {}^3u(\eta_n) - {}^3u_n$ the global error, with ${}^3\epsilon_0 = 0$ for $n = 0$. We get the following error equation

$$\begin{aligned} {}^3\epsilon_{n+1} = {} & \frac{\left(1 - c_0 \frac{h_0 h_n}{h_1}\right)}{\left(1 + c_0 \frac{h_0 h_{n_1}}{h_1}\right)} \, {}^3\epsilon_n - \frac{h_0}{h_1} \frac{1}{\left(1 + c_0 \frac{h_0 h_{n_1}}{h_1}\right)} \\ & \left(\sum_{k=0}^{n-1} {}^3\epsilon_{k+1} (T_{n+\frac{1}{2},k+1} - T_{n+\frac{1}{2},k}) + T_{n+\frac{1}{2},0} \, {}^3\epsilon_0 - T_{n+\frac{1}{2},n} \, {}^3\epsilon_n \right) + {}^3R_{n+\frac{1}{2}}, \end{aligned} \tag{28}$$

where ${}^3R_{n+\frac{1}{2}}$ is the local truncation error of order $O(N^{\alpha-2})$. To show whether the finite difference scheme is convergent, we have to show the error ${}^3\epsilon_{n+1}$ tends to zero as the integration step size $\to 0$. It verifies that for the error equation the following relation holds

$$|\,{}^3\epsilon_{n+1}| \leq K \,|\,{}^3\epsilon_n| + R_{max} \leq K_\alpha \,|\,{}^3\epsilon_n| + R_{max}$$

with K given by (27), $K_\alpha = 1 + c_0 \dfrac{\Delta\eta_{n+1}}{\eta_1^{1-\alpha}}$, and $R_{max} = \max_{0\le n\le N-1} |{}^3R_{n+\frac{1}{2}}|$. The previous formula represents a recursive relation for the global error defined by

$$| {}^3\epsilon_{n+1}| \le a \ | {}^3\epsilon_n| + b$$

where we set $a = K_\alpha \neq 1$ and $b = R_{max}$. Since ${}^3\epsilon_0 = 0$, it easily follows by induction that

$$\begin{aligned} | {}^3\epsilon_{n+1}| &\le (1 + a + a^2 + \cdots + a^n)\ b \\ &\le \frac{a^{n+1} - 1}{a - 1}\ b \\ &= \frac{K_\alpha^{n+1} - 1}{K_\alpha - 1}\ R_{max} \\ &= \frac{\left(1 + c_0 \dfrac{\Delta\eta_{n+1}}{\eta_1^{1-\alpha}}\right)^{n+1} - 1}{c_0 \dfrac{\Delta\eta_{n+1}}{\eta_1^{1-\alpha}}} R_{max}. \end{aligned}$$

Observing that $1 + x < e^x$ with $x > 0$, $(n + 1)\Delta\eta_{n+1} = \tilde{\eta}$ and $\eta_1 = \Delta\eta_1 \le \Delta\eta_{n+1}$, we can write

$$| {}^3\epsilon_{n+1}| \le \frac{e^{c_0 \frac{\tilde{\eta}_{n+1}}{\eta_1^{1-\alpha}}} - 1}{c_0 \dfrac{\eta_1}{\eta_1^{1-\alpha}}} R_{max}.$$

Finally, we have

$$| {}^3\epsilon_{n+1}| \le \bar{K}_\alpha\ R_{max},$$

where $\bar{K}_\alpha = \frac{e^{c_0 \frac{\tilde{\eta}}{\eta_1^{1-\alpha}}} - 1}{c_0 \eta_1^\alpha}$. Then, since $\bar{K}_\alpha$ is a constant independent of the step size, the global error $| {}^3\epsilon_{n+1}| \to 0$ as step size tends to 0 being the method consistent, that is $R_{max} \to 0$ as step size tends to 0. This completes the demonstration of the convergence of the proposed method of order $O(N^{\alpha-2})$.

4. Numerical results and discussion

In this Section, we present the numerical results obtained by solving the mathematical model (15) by the proposed finite difference scheme (25) on the quasi-uniform grid defined by the logarithmic map (16) with control parameter $c = 3$. The tests carried out show the accuracy of the obtained numerical results and the efficiency and reliability of the proposed method. Furthermore, we perform a comparison with the numerical results computed with the classical approach, which consists of the implementing the classical numerical methods, trapezoidal or L1 ones, with the classical truncated boundary. This comparison has highlighted the high accuracy of the numerical solutions obtained by the proposed approach.

The fractional model (15) is transformed into an equivalent FIVP by setting the missing initial condition, ${}^3u(0) = \beta$, and solved by means of the shooting approach.

The shooting method is a classical approach for solving the boundary value problems of integer order but, also fractional one. By the shooting method, the boundary value problem is reduced to an initial value problem and it consists in finding the missing initial value for which the solution of the corresponding IVP satisfies the boundary conditions of the original problem. We set a missing initial guess β and integrate the IVP. We want to find the initial value β such that the numerical solution satisfies the known boundary condition, this value is found by means iterative method as the root of the equation

$$F(\beta) = g(\beta, u(\eta_N, \beta)) = 0.$$

Then, the value of β that satisfies the above equation is an approximation of the missing initial value to which the known boundary value corresponds. The iterative formula used in this study is the secant method. So that, starting

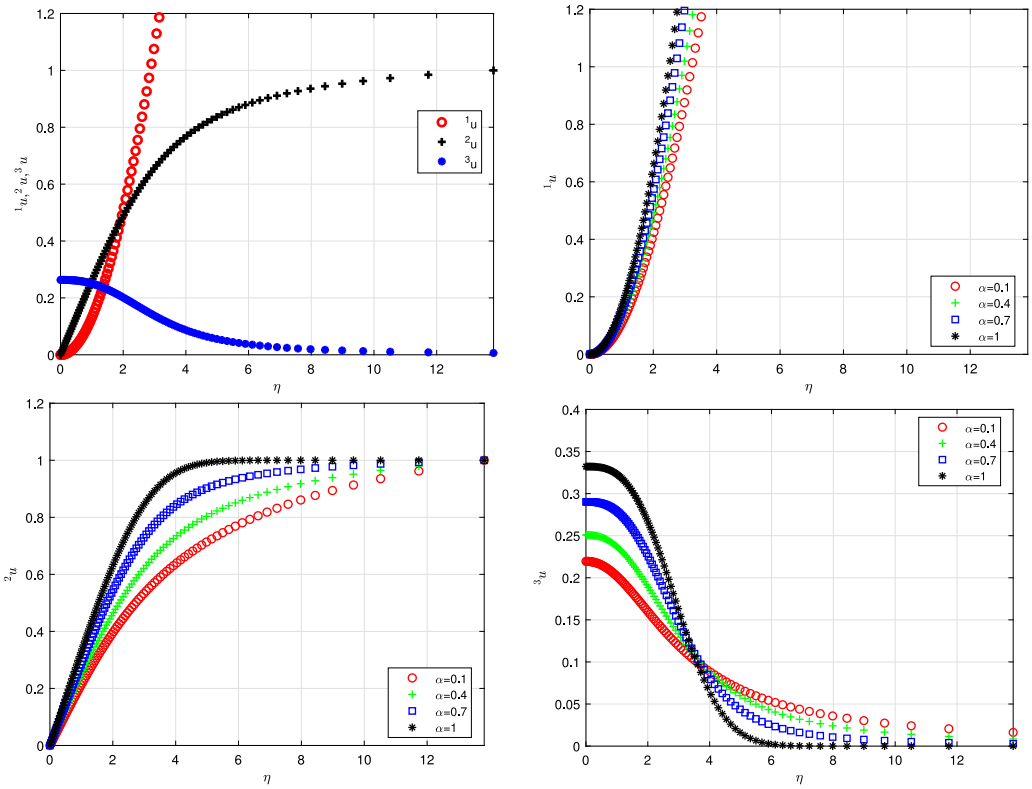

Fig. 4. Numerical solution for the problem (15). Top: Left frame: solution for $\alpha = 0.5$. Right frame: ^1u for different values of α. Bottom: Left frame: ^2u for different values of α. Right frame: ^3u for different values of α.

from two suitable values, β_0 and β_1, the FIVP has been solved numerically by the proposed numerical method and one finds the solution $^3u(\eta_N; \beta)$ as root of the equation

$$F(\beta) = {}^3u_N - 1$$

by a secant root-finding solver with a fixed tolerance TOL $= 10^{-12}$. For the whole range of α, we set as initial values the following

$$\beta_0 = 1, \qquad \beta_1 = 0.1. \tag{29}$$

Note that, it is possible to apply the shooting method also implementing the iterative Newton's method. It is important to note that, from the computational complexity point of view, the Newton method requires the evaluation of the Jacobian, which generally can be computed using the finite differences. Because the sensitivity of solutions of an initial value problem to its initial conditions may be too much to get reasonably accurate derivative values, it is advantageous to compute the Jacobian as a solution to ODEs. This requires a more complex numerical treatment of the model because a system with a doubled dimension has to be solved. For all these reasons, the authors decide to exclude it. Then, we have found that the secant method is particularly suitable in this context.

Fig. 4 shows the numerical solution obtained with $N = 100$ grid points. We report the results on a grid of $N = 100$ grid points in order to show the distribution of the points of the quasi-uniform grid. The problem (12)–(13) is shown to have an initial layer near the initial point $\eta = 0$. The proposed finite difference method implemented on the logarithmic quasi uniform grid reveals an efficient and reliable tool for the problem under study: the distribution of grid points denser at the beginning of the process, permits to approximate the solution more accurately in that part of the domain where the solution exhibits a particular behaviour. The numerical results demonstrate that under the same speed of external flow, the viscoelastic fluids ($0 < \alpha < 1$) move slower than the Newtonian fluid ($\alpha = 1$) in the boundary layer near the plate. The smaller the value of α is, the slower the speed of viscoelastic fluid becomes near the plate.

Table 1
Comparison of the values of ${}^2u$ at point $\eta = 0$ for increasing values of α and $N = 1000$.

α	Method (23) on QUG	TR method on UG	L1 method on UG
0.1	0.191213410931341	0.186214448436116	0.191585263003031
0.2	0.204281463030207	0.195067378684358	0.204845402880093
0.3	0.218659462797643	0.205767110153840	0.219450874993960
0.4	0.234127799578452	0.218329297626890	0.235184393099523
0.5	0.250407273180429	0.232759495172022	0.251768186102331
0.6	0.267169184106165	0.249041926923820	0.268876018125504
0.7	0.284056323497335	0.267140433615328	0.286153633587883
0.8	0.300710618847697	0.287012142557855	0.303247155770632
0.9	0.316802690444417	0.308636384127800	0.319855343918085
1.0	0.332059309843710	0.332056659525770	0.335753345881383

Table 1 shows the initial missing values of the first order derivative $f'(\eta) = {}^2u_0$ for the model (15) for increasing values of α obtained with $N = 1000$. In order to perform a comparison, we report the numerical results obtained by using the proposed numerical method implemented with the quasi-uniform grid and the classical trapezoidal and L1 numerical methods implemented with the truncated boundary on a uniform grid. It is worth to note that the convergence order of the trapezoidal scheme is $O((\Delta\eta)^{min(1+\alpha,2)})$: usually is $1+\alpha$ when $0 < \alpha < 1$ and only when $\alpha > 1$, or when the solution is sufficiently smooth, the reached order 2 is obtained [19,20]. Furthermore, the convergence order of the classical L1 scheme is $O((\Delta\eta)^{2-\alpha})$. By using these two last numerical methods with the truncated boundary, we need to select a suitable large computational domain. So that, in order to perform a comparison, we select the computational domain $[0, 20.7]$, being $\eta_{N-1} = 20.7$ in the quasi uniform grid built with control parameter $c = 3$. All methods are implemented with the well-known shooting approach. In Table 1, we report the results obtained by using the proposed finite difference scheme on the quasi-uniform grid, the numerical values computed by the classical trapezoidal scheme and, in the last column, the numerical values computed by the classical L1 one scheme, respectively. Note that, by the proposed numerical method, we do not need to truncate the domain, because we use the whole domain, being the last grid point located at infinity.

Table 1 displays the comparison of the computed numerical values of ${}^2u$ at point $\eta = 0$ for increasing values of α. The benchmark value found by Blasius for the original problem is ${}^2u(0) = 0.33205733621519630$, here reported in order to compare it with the missing initial condition in Table 1 corresponding to the value $\alpha = 1$.

In this contest, we cannot perform a numerical investigation about the order of accuracy of the numerical method following the classical approach, because we do not know the exact solution of the proposed model and furthermore, due to the structure of the quasi-uniform grid. Then, in order to perform another check on the accuracy of the numerical results, we report the following values of the relative errors

$$Rel\ Err = \frac{\left|{}^2u(0) - {}^2u_{ref}(0)\right|}{\left|{}^2u_{ref}(0)\right|} = 5.943638e-06,$$

$$Rel\ Err = \frac{\left|{}^2u_{TR}(0) - {}^2u_{ref}(0)\right|}{\left|{}^2u_{ref}(0)\right|} = 2.037869e-06,$$

$$Rel\ Err = \frac{\left|{}^2u_{L1}(0) - {}^2u_{ref}(0)\right|}{\left|{}^2u_{ref}(0)\right|} = 1.1130642-02.$$

${}^2u(0)$ is the missing initial condition computed by the proposed method on the quasi uniform grid for $\alpha = 1$. ${}^2u_{TR}(0) = 0.332059309843710$ and ${}^2u_{L1}(0) = 0.335753345881383$ are the missing initial conditions computed by classical trapezoidal method and L1 one, on the uniform grid for $\alpha = 1$, respectively. We consider ${}^2u_{ref}(0) = 0.33205733621519630$ as the reference value computed by Blasius in [4] for $\alpha = 1$ and also in [5,14] The relative errors, reported above, show the accuracy of the obtained results suggesting that the proposed method reveals to be an accurate and efficient scheme for solving also the classical Blasius model of integer order.

Fig. 5 shows the comparison among the numerical solutions for the problem (15) obtained by the proposed method on quasi-uniform grid and classical trapezoidal one with truncated boundary on a uniform grid with $N = 1000$ for $\alpha = 0.1, 0.3, 0.5$ and $\alpha = 1$. The obtained results are in good agreement with the results by the classical approach.

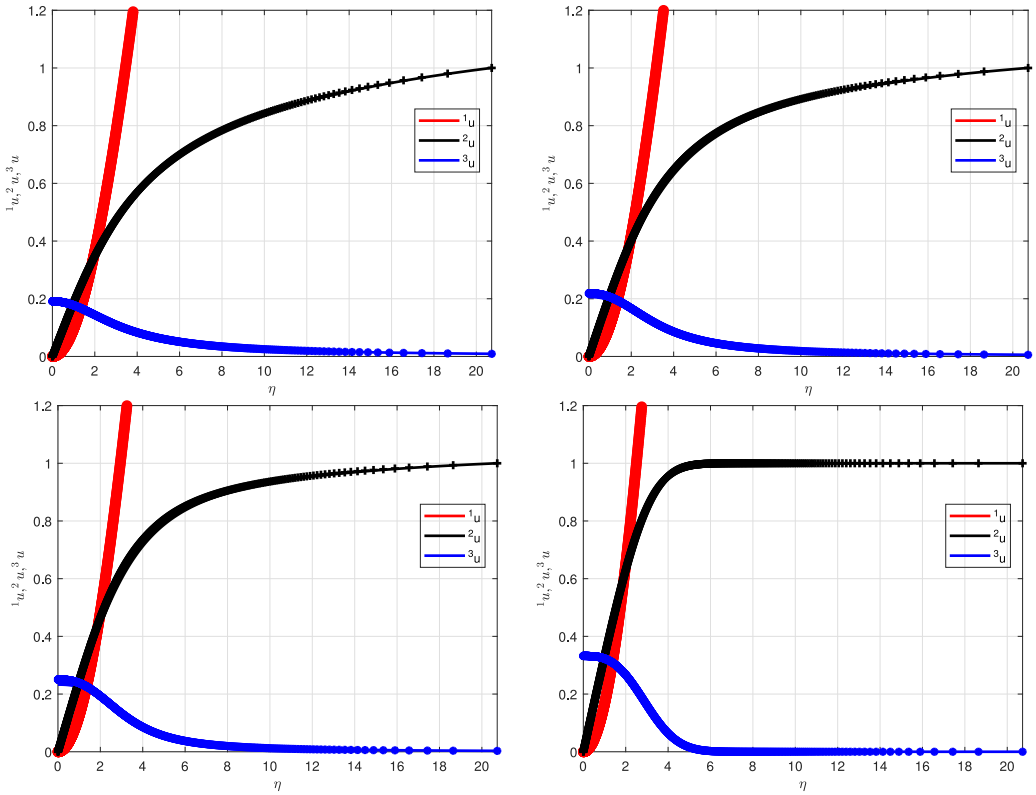

Fig. 5. Comparison of the numerical solutions with $N = 1000$. Top frames: Left frame: $\alpha = 0.1$. Right frame: $\alpha = 0.3$. Bottom frames: Left frame: $\alpha = 0.5$. Right frame: $\alpha = 1$. Continuous lines: solutions by trapezoidal method; Dots lines: solutions by (23) on QUG.

5. Concluding remarks

In this paper, we propose a fractional mathematical model for the Blasius flow described by a non-linear two-point boundary value problem on a semi-infinite interval in terms of the Caputo derivative. By introducing a suitable discretization of the computational domain, constructed in such a way that the boundary conditions at infinity are exactly assigned, we propose an efficient finite difference scheme where the Caputo fractional derivative is approximated by means of a suitable approximation formula. We discuss the consistency, stability and convergence of our proposed scheme. The numerical results are reported. We conclude that the proposed model represents a good modelization for the fractional problem that reduces to the classical Blasius model of integer order for α towards one. Furthermore, the proposed finite difference method, implemented on the logarithmic quasi-uniform grid, reveals to be an efficient and reliable approach for capturing the behaviour boundary layer of the solution at the initial point.

Finally, we remark that the proposed method, developed for solving the fractional Blasius problem, can be used to solve the class of FBVPs defined on infinite or semi-infinite interval.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets generated and analysed during the current study are available from the corresponding author on reasonable request.

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